

Where Does the Cache Hit Ratio Go?

Lian Zhou, Ruoxi Feng, Kaiyuan Wang

title: "Where Does the Cache Hit Ratio Go? An Empirical Decomposition of Edge CDN Workloads"

authors: "Lian Zhou, Ruoxi Feng, Kaiyuan Wang"

Abstract

Edge content delivery networks report aggregate cache hit ratios as their primary efficiency metric, yet practitioners routinely observe that two deployments with the same aggregate ratio deliver very different end-to-end latency. We conduct a measurement study over four weeks of production traces from a regional edge deployment serving 63 million requests. We decompose the aggregate hit ratio by content class, by request-size bucket, and by delivery path, and we quantify how much of the variance in origin offload is explained by each decomposition. Our central finding is that byte-level hit ratio and request-level hit ratio diverge sharply: video segments account for 7.1 percent of requests but 68.4 percent of bytes, so a deployment optimized for request count systematically misestimates origin egress. We release the aggregated workload statistics used in this paper.

1 Introduction

Caching is the single most effective lever for reducing origin load in content delivery. Operators therefore monitor cache hit ratio closely, and it appears in almost every vendor's dashboard. The metric is appealingly simple, but it hides a well-known ambiguity: it can be computed over requests or over bytes, and the two answers can differ by an order of magnitude.

This paper asks a narrow, operational question: if we decompose the aggregate hit ratio along the dimensions that operators can actually control, which decomposition best predicts origin egress and tail latency? Answering it requires request-level traces rather than dashboard aggregates, so we instrumented a regional edge deployment for four weeks.

Our contributions are threefold. First, we provide a systematic decomposition of cache performance across content class, object size, and delivery path. Second, we quantify the mismatch between request-level and byte-level accounting. Third, we show that a two-parameter model over content class explains most of the observed variance in origin offload.

2 Related Work

Prior measurement studies of web caches focused on request-level hit ratios and on the effect of replacement policies under synthetic workloads. Trace-driven simulation work established that object size distributions are heavy-tailed, which implies that request-weighted and byte-weighted metrics must diverge. Our study confirms this on a modern edge deployment and extends it with a delivery-path decomposition that earlier work did not have access to.

3 Methodology

3.1 Measurement Setup

We collected four weeks of access logs from a regional edge deployment with nine points of presence. Each log record contains a timestamp, a content class label assigned by the origin's content management system, the response size in bytes, the delivery path (edge hit, parent hit, or origin fetch), and the time to last byte. We excluded requests smaller than 100 bytes because they are dominated by control-plane traffic.

3.2 Metrics

We define the request-level hit ratio as the fraction of requests served without contacting the origin, and the byte-level hit ratio as the fraction of response bytes that were served from cache. We also report origin egress reduction, which is one minus the ratio of origin-served bytes to total bytes.

4 Results

4.1 Traffic Composition

Table 1 reports the composition of the workload by content class.

Content class	Request share (%)	Byte share (%)	Request hit ratio	Byte hit ratio
Static assets	46.8%	12.1%	0.94	0.91
Images	27.4%	12.4%	0.88	0.84
Video segments	7.1%	68.4%	0.62	0.58
API responses	18.7%	7.1%	0.41	0.39

Table 1. Workload composition and cache performance by content class. Request share and byte share sum to 100 percent.

4.2 Aggregate Totals

Table 2 aggregates the deployment over the whole measurement window.

Metric	Value
Total requests	63,214,880
Total bytes	412,660,000,000
Request hit ratio	0.83
Byte hit ratio	0.61
Origin egress reduction	0.61
Average time to last byte	148.0

Table 2. Aggregate deployment statistics over the four week window.

4.3 Why Request Share and Byte Share Diverge

Video segments are the extreme case. They represent 7.1 percent of requests but 68.4 percent of bytes, and their byte hit ratio of 0.58 is the second lowest observed. Consequently the region's aggregate byte hit ratio, 0.61, is more than

twenty points below its request hit ratio of 0.83.

Figure 1. Request share versus byte share by content class. The two distributions disagree most sharply for video segments.

4.4 A Two-Parameter Model

Fitting origin egress as a function of content-class request share and class byte hit ratio yields a two-parameter model that explains 0.93 of the variance across the nine points of presence.

5 Discussion

Operators who monitor only request-level hit ratios will under-provision origin bandwidth whenever video share grows. We recommend reporting both metrics side by side, and we recommend weighting capacity planning by byte share rather than request share.

6 Conclusion

Aggregate cache hit ratio is an ambiguous metric. Decomposing it by content class reveals that byte-level and request-level accounting can differ by more than twenty percentage points in a realistic edge deployment. Our two-parameter model over content class explains most of the variance in origin offload and is cheap to instrument.

7 Data Availability

The aggregated workload statistics underlying Table 1 and Table 2 are provided alongside this paper as a CSV file. Raw request logs cannot be released because they contain tenant-identifying fields.